

BOULIA, Australia

WMO: 943330

Lat: 22.912S

Lon: 139.904E

Elev: 519

StdP: 14.42

Time Zone: 10.00 (E10)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.3	46.4	13.5	11.1	66.8	17.5	13.6	64.2	19.6	66.5	17.9	67.3	4.2	180	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	106.7	71.8	104.7	71.7	102.8	71.4	79.9	94.7	78.3	91.8	77.2	90.5	6.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	142.2	85.0	75.3	135.3	83.3	73.8	128.6	83.6	44.0	95.0	42.3	91.4	41.1	90.2	91.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.3	17.1	DB	39.3	110.0	1.9	1.6	38.0	111.1	36.8	112.0	35.8	112.9	34.4	114.0	(4)
(5)			WB	33.5	85.1	2.4	2.5	31.8	86.9	30.4	88.4	29.0	89.8	27.3	91.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.8	88.8	88.7	84.8	77.4	68.0	62.8	61.7	65.1	74.1	79.2	84.0	87.2	(6)
(7)		DBStd	11.40	5.28	4.42	4.74	6.15	5.60	6.28	5.68	6.06	6.91	6.25	6.01	5.40	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	375	0	0	0	3	34	117	136	73	11	1	0	0	(9)
(10)		CDD50	9767	1203	1083	1080	821	559	384	364	470	724	905	1019	1155	(10)
(11)		CDD65	4669	738	663	615	374	128	52	36	78	285	441	569	690	(11)
(12)		CDH74	64734	11377	10008	8533	4212	1115	458	376	987	3505	5781	8053	10329	(12)
(13)		CDH80	37605	7347	6293	4890	2039	285	95	60	313	1732	3185	4840	6526	(13)
(14)	Wind	WSAvg	6.2	5.8	6.0	6.0	5.2	4.8	4.8	4.8	6.4	7.1	8.0	7.6	7.4	(14)
(15)	Precipitation	PrecAvg	10.2	2.1	1.9	1.1	0.6	0.6	0.3	0.4	0.2	0.2	0.6	0.9	1.4	(15)
(16)		PrecMax	30.5	18.3	9.0	10.0	4.6	4.2	2.4	4.6	1.5	1.6	3.3	5.9	10.2	(16)
(17)		PrecMin	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	6.6	3.0	2.4	1.7	1.1	0.9	0.6	0.8	0.4	0.4	0.8	1.3	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	108.9	107.0	104.0	99.4	88.9	86.2	85.1	91.1	100.0	104.8	107.1	109.2	(19)
(20)			MCWB	72.3	73.0	69.5	70.4	62.9	65.3	59.8	64.1	66.5	70.2	70.3	70.2	(20)
(21)		2%	DB	106.1	104.5	101.2	95.8	86.4	82.9	80.7	87.3	95.8	101.1	104.6	106.5	(21)
(22)			MCWB	72.6	73.2	70.1	68.2	63.3	63.8	59.3	62.6	66.2	68.8	70.5	71.4	(22)
(23)		5%	DB	103.8	102.4	99.1	93.3	84.0	79.9	78.1	84.0	93.3	97.9	102.0	104.1	(23)
(24)			MCWB	71.9	72.8	70.0	67.7	62.8	61.6	58.1	60.6	65.3	67.0	69.9	71.7	(24)
(25)		10%	DB	101.5	100.1	96.6	90.7	80.9	75.7	75.0	80.2	90.4	95.0	98.8	101.2	(25)
(26)			MCWB	71.6	72.5	69.5	65.9	61.5	59.1	56.1	58.2	63.4	65.3	69.1	70.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.4	82.5	79.0	78.0	69.8	68.2	65.9	68.9	73.8	77.9	81.6	81.0	(27)
(28)			MCD B	98.6	98.1	87.6	90.9	77.2	82.4	77.7	83.3	94.0	100.1	95.8	101.4	(28)
(29)		2%	WB	79.3	79.5	76.8	73.6	66.9	65.5	62.4	65.7	70.8	73.5	76.3	78.3	(29)
(30)			MCD B	93.9	92.1	86.7	85.2	79.2	79.8	76.9	82.4	88.5	91.1	92.9	94.1	(30)
(31)		5%	WB	77.5	78.1	75.0	71.5	65.1	63.1	59.7	62.7	68.3	70.8	74.4	76.9	(31)
(32)			MCD B	91.8	90.4	87.3	86.0	78.6	76.8	75.5	79.6	87.1	87.7	89.4	90.3	(32)
(33)		10%	WB	76.2	77.2	73.3	69.4	63.3	60.9	57.5	60.1	65.8	68.6	72.8	75.5	(33)
(34)			MCD B	91.1	89.4	89.0	85.4	77.4	73.7	73.0	77.9	84.8	86.9	88.2	89.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.1	20.7	20.9	21.3	21.0	21.8	23.5	24.9	25.7	24.9	23.7	22.1	(35)
(36)			MCDBR	23.9	23.1	23.5	24.3	23.5	24.4	26.5	28.3	28.6	28.1	28.0	27.3	(36)
(37)			MCWBR	8.6	9.3	9.5	10.6	11.0	11.8	13.9	14.3	13.4	12.5	12.4	11.9	(37)
(38)		5% WB	MCDBR	20.0	18.6	17.2	20.1	18.7	21.8	22.6	25.1	25.0	22.3	22.3	19.7	(38)
(39)	MCWBR		8.2	6.7	7.4	11.1	10.7	10.9	12.9	13.5	13.6	11.0	10.8	9.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.383	0.376	0.357	0.340	0.326	0.310	0.309	0.324	0.369	0.374	0.388	0.393	(40)	
(41)		taud	2.436	2.454	2.478	2.480	2.470	2.497	2.472	2.406	2.306	2.314	2.331	2.362	(41)	
(42)		Ebn at Noon	305	303	300	292	283	282	286	292	291	300	301	302	(42)	
(43)		Edh at Noon	39	38	35	32	30	28	30	34	41	43	43	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2371	2345	2117	1855	1512	1347	1461	1783	2084	2356	2427	2402	(44)	
(45)		RadStd	171	143	141	132	82	80	85	73	144	159	119	178	178	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page